

**PH3244**  
**Experiment - 1**  
**Measurement of Suseptibility of a liquid or a solution by**  
**Quincke's method**

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**Synopsis**

In this experiment we try to measure the suseptibility of given substance by Quincke's method.

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## I. THEORY AND PROCEDURE

### A. Theory

The Quincke's method is used to determine magnetic susceptibility of diamagnetic or paramagnetic substances in the form of a liquid or an aqueous solution. When an object is placed in a magnetic field, a magnetic moment is induced in it. Magnetic susceptibility  $\chi$  is the ratio of the magnetization  $I$  (magnetic moment per unit volume) to the applied magnetizing field intensity  $H$ . The magnetic moment can be measured either by force methods, which involve the measurement of the force exerted on the sample by an inhomogeneous magnetic field or induction methods where the voltage induced in an electrical circuit is measured by varying magnetic moment. The Quincke's method like the Gouy's method belongs to the former class. The force  $f$  on the sample is negative of the gradient of the change in energy density when the sample is placed,

$$f = \frac{d}{dx} \left[ \frac{1}{2} \mu_0 (\mu_r - \mu_{ra}) H^2 \right] = \frac{1}{2} \mu_0 (\chi - \chi_a) \frac{d}{dx} H^2 \quad (1)$$

Here  $\mu_0$  is permeability of the free space and  $\mu_r, \chi$  and  $\mu_{ra}, \chi_a$  are respectively relative permeability and susceptibility of the sample and the air which the sample displaces. The force acting on an element of area  $A$  and length  $dx$  of the liquid column is  $fAdx$ , so the total force  $F$  on the liquid is

$$F = A \int f dx = \frac{A\mu_0}{2} (\chi - \chi_a) (H^2 - H_0^2) \quad (2)$$

where the integral is taken over the whole liquid. This means that  $H$  is equal to the field at the liquid surface between the poles of the magnet and  $H_0$  is the field at the other surface away from the magnet. The liquid (density  $\rho$ ) moves under the action of this force until it is balanced by the pressure exerted over the area  $A$  due to a height difference  $h$  between the liquid surfaces in the two arms of the U-tube. It follows that

$$F = Ah(\rho - \rho_a)g$$

Or

$$\chi = \chi_a + \frac{2}{\mu_0} g(\rho - \rho_a) \frac{h}{(H^2 - H_0^2)} \quad (3)$$

In actual practice  $\chi_a$ , density of air  $\rho_a$  and  $H_0$  are negligible and can be ignored and the above expression simplifies to

$$\chi = \frac{2\rho gh}{\mu_0 H^2} \quad (4)$$

This equation shows that by plotting  $h$  as a function of  $H^2$ , the susceptibility  $\chi$  can be determined from the slope. The expressions in C.G.S units are given by

$$\chi = \chi_a + 2g(\rho - \rho_a) \frac{h}{(H^2 - H_0^2)} \quad (5)$$

$$\boxed{\chi = \frac{2\rho gh}{H^2}} \quad (6)$$

The mass susceptibility  $\chi'$  is given by  $\chi/\rho$  and the molar susceptibility  $\chi''$  by  $M\chi'$  where  $M$  is the molecular weight of the specimen. In the case of solutions, correction must be made for the diamagnetic contribution of water. If the number of water molecules per unit volume is not very different in the solution from that in pure water,

$$\boxed{\chi_{\text{solution}} = \chi_{\text{salt}} + \chi_{\text{water}}} \quad (7)$$

In general if  $m_s$  is the mass of the salt dissolved in  $m_w$  of water,

$$\boxed{\chi'_{\text{solution}} = \frac{m_s}{m_s + m_w} \chi'_{\text{salt}} + \frac{m_w}{m_s + m_w} \chi'_{\text{water}}} \quad (8)$$

Mass susceptibility of the salt,  $\chi'_{\text{salt}}$  can be obtained from this relation.

In a paramagnetic substance there are non-interacting permanent magnetic dipoles. The magnetizing field tends to align these parallel to the field. Thermal effects on the other hand tend to destroy this alignment. The result is that, for  $kT \gg \mu_0 \mu_B H$ , the volume susceptibility  $\chi_{\text{salt}}$  at temperature  $T$  is given by

$$\chi_{\text{salt}} = \frac{I}{H} = \frac{N \mu_0 (p \mu_B)^2}{3kT} = \frac{C}{T}, \quad (\text{S.I. units}) \quad (9)$$

$$\chi_{\text{salt}} = \frac{N (p \mu_B)^2}{3kT} = \frac{C}{T}, \quad (\text{C.G.S. units}) \quad (10)$$

where  $k$  is Boltzmann constant,  $N (= N_A \rho_{\text{salt}}/M)$  is the number of ions per unit volume and  $\mu_B$  is Bohr magneton.  $p$  is the magneton number which may be calculated from the measured value of  $\chi_{\text{salt}}$  (or  $\chi'_{\text{salt}} = \chi_{\text{salt}}/\rho_{\text{salt}}$ ) and compared with the theoretical value depending on  $L$ ,  $S$  and  $J$  of the magnetic ion as given in Appendix I. Eq. 10 expresses Curie law and the constant  $C$  is called the Curie constant. The above expression assumes that the magnetic field acting on each ion is just the applied field and contributions due to neighbouring magnetic ions are neglected. For dilute paramagnetic materials this approximation is valid.

## B. Procedure

A schematic diagram of Quincke's set up is shown in Fig.1. One limb of the glass U-tube is very narrow (about 2 – 3 mm in diameter) and the other one quite wide. The result is that a change in the level of the liquid in the narrow limb does not affect the level in the wider limb. The narrow limb is placed between the pole pieces of an electromagnet shown as N-S such that the meniscus of the liquid lies symmetrically between N-S. The length of the limb should be sufficient enough to keep the lower extreme end of this limb well outside the field of the magnet. The rise or fall  $h$  on applying the field is measured by means of a traveling microscope fitted with a micrometer scale of least count of order  $10^{-3}$  cm.

### 1. Measurement of $H$

1. Fix the air gap between the pole pieces of the electromagnet to the minimum distance required to insert Quinck's tube without touching the pole pieces.
2. Measure the air gap. Each time the air gap changes, the graph will change.
3. Mount the Hall probe of the Digital Gaussmeter, DGM-102 in the wooden stand provided and place it at the centre of the air gap such that the surface of the probe is parallel to the pole pieces. The small black crystal in the probe is its transducer, so this part should be at the centre of the air gap.

4. Connect the leads of the Electromagnet to the Power Supply, bring the current potentiometer of the Power Supply to the minimum. Switch on the Power Supply and the Gaussmeter.
5. Slowly raise the current in the Power Supply and note the magnetic field reading in the Gaussmeter.
6. Plot the graph between the current and the magnetic field. This graph will be linear for small values of the current and then the slope will decrease as magnetic saturation occurs in the material of the pole pieces.

## 2. Measurement of $h$

1. Test and ensure that each unit (Electromagnet and Power Supply) is functioning properly.
2. Measure the density  $\rho$  of the specimen (liquid or solution) by specific gravity bottle. If the mass of empty bottle is  $w_1$ , filled with specimen  $w_2$  and filled with water  $w_3$ , then

$$\rho = \rho_{\text{water}} \frac{w_2 - w_1}{w_3 - w_1} \quad (11)$$

3. *Scrupulous* cleaning of the tube is essential. *Thoroughly* clean the Quincke's tube, rinse it well with distilled water and dry it (preferably with dry compressed air). Do not use the tube for longer than one laboratory period without recleaning it.
4. Keep the Quincke's tube between the pole pieces of the magnet as shown in Fig.1. The length of the horizontal connecting limb should be sufficient to keep the wide limb out of the magnetic field.
5. Fill the liquid in the tube and set the meniscus centrally within the pole pieces as shown. Focus the microscope on the meniscus and take reading.
6. Apply the magnetic field  $H$  and note its value from the calibration, which is done earlier as an auxiliary experiment. Note whether the meniscus rises up or descends down. It rises up for paramagnetic liquids and solutions while descends down for diamagnetics. Readjust the microscope on the meniscus and take reading. The difference of these two readings gives  $h$  for the field  $H$ . The magnetic field between the poles of the magnet does not drop to zero even when the current is switched off. There is a residual magnetic field  $R$  which requires a correction.
7. Measure the displacement  $h$  as a function of applied field  $H$  by changing the magnet current in small steps. Plot a graph of  $h$  as a function of  $H^2$ .

## II. OBSERVATION

### A. Measurement of $h$ and $H$

| Current (A) | Main Scale (mm) | Circular Scale (0.01 mm) |
|-------------|-----------------|--------------------------|
| 0.00        | 9.5             | 48                       |
| 0.80        | 11.5            | 10                       |
| 1.00        | 12.5            | 5                        |
| 1.25        | 13.5            | 3                        |
| 1.50        | 16.5            | 2                        |
| 1.75        | 18.5            | 12                       |
| 2.00        | 20.0            | 19                       |
| 2.50        | 24.0            | 12                       |
| 2.75        | 24.0            | 38                       |
| 3.00        | 24.5            | 15                       |
| 3.25        | 25.0            | 40                       |
| 3.30        | 25.0            | 48                       |

TABLE I:  $\Delta h$  measurement; Data taken on 21/01/26

| Current (A) | Field Strength (Gauss) |
|-------------|------------------------|
| 0.00        | 423                    |
| 0.10        | 729                    |
| 0.20        | 1052                   |
| 0.30        | 1400                   |
| 0.40        | 1769                   |
| 0.50        | 2150                   |
| 0.60        | 2550                   |
| 0.70        | 2970                   |
| 0.80        | 3380                   |
| 0.90        | 3790                   |
| 1.00        | 4210                   |
| 1.50        | 6340                   |
| 2.10        | 8490                   |
| 2.50        | 10230                  |
| 3.00        | 11840                  |
| 3.50        | 13260                  |
| 3.55        | 13390                  |

TABLE II:  $H$  measurement; Data taken on 21/01/26

Mass and weight data not taken, experiment repeated and the readings are as follows:

| Current (A) | H (10*Gauss) | Main Scale (mm) | Circular Scale (0.01 mm) |
|-------------|--------------|-----------------|--------------------------|
| 0.00        | 64           | 10              | 46                       |
| 0.50        | 277          | 10.5            | 45                       |
| 0.75        | 405          | 11              | 9                        |
| 1.00        | 530          | 11              | 36                       |
| 1.20        | 625          | 12.5            | 35                       |
| 1.37        | 705          | 15              | 10                       |
| 1.58        | 812          | 16              | 35                       |
| 1.81        | 915          | 17.5            | 11                       |
| 2.01        | 1001         | 21.5            | 20                       |
| 2.20        | 1083         | 23.5            | 0                        |

TABLE III: Simutanious measurement of both  $H$  and  $h$ ; Data taken on 16/02/26

Mass of solute: 8.4933 g  
Volume of water: 17.0 ml  
Final volume of water: 20.0 ml

Least count of readings are the significant figures written in the tables. add 3M, conc and other data, also add the previous data too of that table to not include Cus04 ka data, why repeat add and toehr stuff toto

III. CALCULATION AND ANALYSIS

A. Sources of Error

B. Error propagation

C. Calculations

From Table - 3 we calculate  $h$

| Current (A) | H (10*Gauss) | Main Scale (mm) | Circular Scale (0.01 mm) | $h$ (mm) | $H^2$ (G) |
|-------------|--------------|-----------------|--------------------------|----------|-----------|
| 0           | 64           | 10              | 46                       | 0        | 409600    |
| 0.5         | 277          | 10.5            | 45                       | 0.49     | 7672900   |
| 0.75        | 405          | 11              | 9                        | 0.63     | 16402500  |
| 1           | 530          | 11              | 36                       | 0.9      | 28090000  |
| 1.2         | 625          | 12.5            | 35                       | 2.39     | 39062500  |
| 1.37        | 705          | 15              | 10                       | 4.64     | 49702500  |
| 1.58        | 812          | 16              | 35                       | 5.89     | 65934400  |
| 1.81        | 915          | 17.5            | 11                       | 7.15     | 83722500  |
| 2.01        | 1001         | 21.5            | 20                       | 11.24    | 100200100 |
| 2.2         | 1083         | 23.5            | 0                        | 13.04    | 117288900 |

TABLE IV: Simutanious measurement of both  $H$  and  $h$ ; Data taken on 16/02/26

Plotting the data from Table - 4 we get:

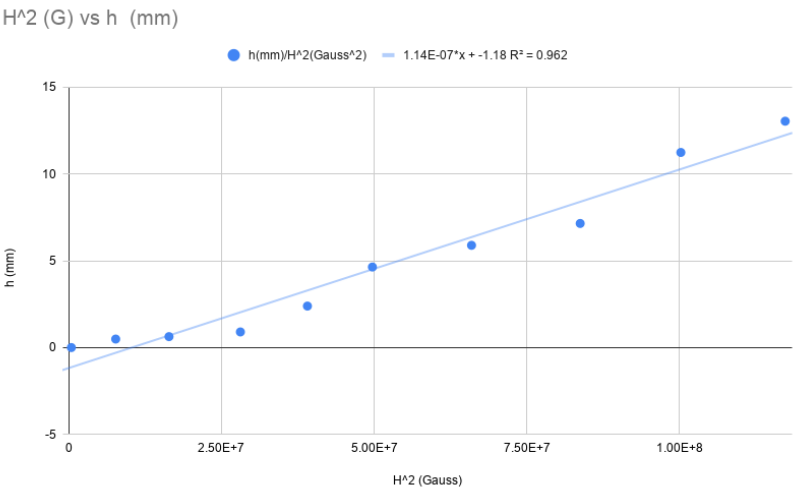


FIG. 1: Plot of Table - 4 with a linear fit and no error bars

from this we get a slope of . using ..

IV. RESULT AND CONCLUSION

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## REFERENCES